Meta-ARA: The Dark Sector Formula

Three Coupled Pairs and the Cosmic Energy Budget

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Scripts 243BL – 243BL8

The Formula

From five axioms and one constant (φ), we derive the entire cosmic energy budget to sub-percent accuracy.

Axioms

1. **Flatness:** $\Omega_{total} = 1$ (the universe is spatially flat)
2. **Horizontal coupling:** DE/DM = φ^2 (Space↔Time coupler from the three-circle architecture)
3. **Diagonal coupling:** DM/baryons = $\varphi^{7/2}$ (crossing two singularities on the four-system grid)
4. **Three coupled pairs:** Space/Time, Light/Dark, Information/Matter
5. **Self-similarity:** same architecture at every scale

Derivation

From axioms 1–3, the three-component constraint system is:

$$DE + DM + b = 1, \quad DE = \varphi^2 \times DM, \quad b = DM / \varphi^{7/2}$$

Substituting into the flatness condition:

$$DM \times (\varphi^2 + 1 + \varphi^{-7/2}) = 1$$

$$\mathbf{DM = 1 / (\varphi^2 + 1 + \varphi^{-7/2}) = 0.2629}$$

Results

Component	Predicted	Observed (Planck)	Δ
Ω_{de} (Dark Energy)	0.6883	0.6850	0.48%
Ω_{dm} (Dark Matter)	0.2629	0.2650	0.79%
Ω_b (Baryons)	0.0488	0.0493	1.03%
Ω_m (Total Matter)	0.3117	0.3143	0.83%

Average error: 0.77%. The entire cosmic energy budget from the golden ratio and two structural rules.

Additional Predictions

Prediction	Predicted	Observed	Δ
Hubble tension	H_0(local) = 73.48	73.0 ± 1.0	0.7%
Transition redshift	$z = 1/\varphi = 0.618$	0.632	2.3%
ARA_space	$\varphi^{-\varphi} = 0.4590$	0.4588	0.04%
DM/baryons	$\varphi^{3.5} = 5.389$	5.375	0.2%
DE/DM	$\varphi^2 = 2.618$	2.585	1.3%

The Three Coupled Pairs

The universe is structured by three fundamental paired axes. Each pair has two poles and a singularity at its centre where the poles become indistinguishable. These three pairs form their own three-circle ARA system — a meta-ARA that is self-similar with the micro-level architecture.

Pair 1: Space ↔ Time

The stage on which everything plays out. Dark Energy lives in the Time pole (it stretches light via redshift). Dark Matter lives in the Space pole (it bends light via gravitational lensing). The coupling between these poles is φ^2 — the horizontal coupler from the three-circle architecture. The singularity S_1 is the Big Bang, where space and time are indistinguishable.

ARA = φ^(−φ) = 0.459 — consumer. Space is being consumed by expansion. This is a self-referential value: the base and exponent are the same number.

Pair 2: Light ↔ Dark

The visibility axis. The dark sector (DE + DM = 95%) vastly dominates the visible sector (baryons + radiation = 5%). The singularity S_2 is the event horizon — cosmologically manifest as the surface of last scattering (CMB at $z \approx 1100$), the boundary between the opaque and transparent universe.

ARA = φ^6.1 ≈ 19.2 — massively dominant. Darkness overwhelms light by nearly 20:1.

Pair 3: Information ↔ Matter

The substance axis. Information is pattern and structure (Dark Matter as gravitational scaffolding). Matter is material realisation (baryonic matter as stars, chemistry, life). The singularity S_3 is the measurement event — where wave becomes particle, where possibility becomes actuality. Cosmologically: structure formation, where information encoded in DM scaffolding becomes material galaxies.

ARA = φ^3.5 = 5.38 — engine. Information outweighs matter. This IS the 3.5 exponent — the ARA of the third pair.

Cosmic Components as Triple Intersections

Each observable cosmic component sits at the intersection of three pair-poles. This is exactly the “beeswax” concept from the original three-circle model: things that exist at the intersection of all three circles are rare and specific.

Component	Space/Time	Light/Dark	Info/Matter
Dark Energy (Ω=0.685)	Time	Dark	Energy (pre-info)
Dark Matter (Ω=0.265)	Space	Dark	Information
Baryons (Ω=0.049)	Space	Light	Matter
Radiation ($\Omega \approx$5e-5)	Time	Light	Energy

Meta-Intersections

The three pairs also produce pairwise intersections that map to the three pillars of modern physics:

- **Space/Time ∩ Light/Dark = Physics** (the 2×2 grid of DE, DM, baryons, radiation — general relativity and cosmology)
- **Light/Dark ∩ Info/Matter = Measurement** (observation, wavefunction collapse, the act of seeing)
- **Space/Time ∩ Info/Matter = Quantum Mechanics** (wave-particle duality played out on the spacetime stage)
- **All three = Observed Reality** (us, here, now — the triple intersection)

Why 7/2 = 3.5

The exponent 3.5 appears throughout the dark sector: DM/baryons = $\varphi^{3.5}$ (0.2% match), the best-fit diagonal coupling in the coupled formula, and the ARA of the Information/Matter pair. It has three convergent derivations:

1. Golden Angle Overshoot

When n systems are placed at golden angle intervals ($360°/\varphi^2 \approx 137.5°$), they overshoot 360° by a factor of $(2n-1) - n\varphi$. For n = 4 systems (the four cosmic components), this gives an overshoot coefficient of $7 - 4\varphi$. The number 7 appears naturally from 4 golden angles. The coupling exponent follows the pattern $(2n-1)/2$, giving 7/2 = 3.5 for n = 4.

2. Additive Coupling (Manhattan Distance)

The 2×2 grid has two coupling axes. The horizontal (Space↔Time) carries exponent 2. The vertical (Dark↔Light) carries exponent 1.5 = $(2\times2-1)/2$ for n = 2 systems. The diagonal coupling = horizontal + vertical = 2 + 1.5 = 3.5. This is a Manhattan distance in coupling space — you cannot take a shortcut through the singularity, you must traverse both axes.

3. Pipe Capacity + Singularity Crossings

$3.5 = 2\varphi + (7-4\varphi)/2$. The pipe capacity $2\varphi = 3.236$ is the "smooth" coupling through the system. The remainder $(7-4\varphi)/2 = 0.264$ is the cost of crossing two singularities (S_2: Dark→Light and S_3: Information→Matter). Each crossing costs $(7-4\varphi)/4$ — one system's share of the 4-system golden angle overshoot. The path from DM to baryons stays in Space (no S_1 crossing) but must cross through the event horizon and the measurement singularity.

Information³ in the Dark Sector

The Information³ framework (Datum → Signal → Meaning) maps directly onto the dark sector:

- **Datum = Dark Energy** — raw, undifferentiated expansion. Vast and formless.
- **Signal = Dark Matter** — gravitational scaffolding. Structure without visibility.
- **Meaning = Baryonic Matter** — stars, chemistry, life. Structure made visible and interpreted.

The coupling chain: DE →(÷φ^2)→ DM →(÷$\varphi^{1.5}$)→ Baryons. The horizontal step (Datum→Signal) is clean at 1.3% because it stays on the same rung. The diagonal step (Signal→Meaning) crosses scales and is structurally messier — but it is *structured* messiness ($\varphi^{3.5}$, not random), exactly as Information³ predicts: datum→signal is filtering, signal→meaning requires interpretation.

Light couples all three levels. It IS information in transit. Dark Matter bends light in SPACE (gravitational lensing). Dark Energy stretches light in TIME (cosmological redshift). This is the flip: the two dark components interact with light along the two different poles of Pair 1.

Self-Similarity

The meta-ARA uses the same couplers as the micro-architecture. At the micro-level: Space, Time, and Rationality are three circles with φ^2 horizontal coupling and $2/\varphi$ vertical coupling. At the meta-level: the three PAIRS are three circles with the same coupling structure. The universe's architecture is fractal in φ.

The three pair-ARA exponents ($-\varphi$, 3.5, 6.1) divide their span by an octave. The left span (Pair 1 to Pair 3, exponents -1.618 to $3.5 = 5.118$) and the right span (Pair 3 to Pair 2, 3.5 to $6.1 = 2.600$) sit in a ratio of 1.97 — an octave (×2), only 1.6% from exact. A $1/\varphi$ reading is 21% off and was rejected. This matches the corrected framework: rung SPACING is octave (the tower), while φ is the coupling/handover BETWEEN rungs (the breathing gap). The dark-sector ratios DE/DM = φ^2 and DM/b = $\varphi^{3.5}$ are couplings, correctly φ; the stacking of the three pairs is spacing, correctly octave.

What Did Not Work

Honesty requires documenting failures alongside successes:

- **The vertical coupler 2/φ as a direct density ratio between dark and light sectors:** Every approach (direct ratio, fraction, weight) missed by hundreds of percent. The dark/light split is 95%/5%, nowhere near 2/φ.
- **Uniform 1/φ reverberation decay:** The three-bounce model with 1/φ decay per step does not reproduce DE→DM→baryons. The steps use different couplers (φ^2 horizontal, φ^3.5 diagonal).
- **Tensor product factorisation of the 2×2 grid:** The grid does NOT factor as (Space/Time) ⊗ (Dark/Light). The axes are entangled. DM/b ≠ DE/γ, confirming the coupling is genuinely non-separable.
- **z_eq/z_CMB = φ^2:** The ratio of matter-radiation equality to CMB decoupling redshifts is 3.12, not φ^2 = 2.62 (19% off). It is closer to π (0.6% off), which is interesting but not the predicted coupling.
- **Signal/Meaning era duration ratio = φ:** The ratio of matter era to current era duration is 1.26, not φ = 1.62 (22% off).
- **Splitting Ω_m into DM and baryons via φ^2:** Using DM = Ω_m × $\varphi^2/(1+\varphi^2)$ gives 73–79% errors on baryons. The φ^2 split works within the dark sector but not for the dark-to-light step.

Testable Predictions

6. **DM/DE ratio constant across redshift** — testable with DESI DR2. The framework predicts the φ^2 coupling holds at all epochs.
7. **Hubble tension resolves to exactly $1/\varphi^5$** — currently 0.7% off observed. Future H_0 convergence should approach 73.5 km/s/Mpc from the local side.
8. **Transition redshift refines to 1/φ** — currently z_trans = 0.632 vs predicted 0.618. DESI + Euclid data will sharpen this.
9. **Dark energy clusters with dark matter** — the framework treats them as coupled on the same plane, predicting DE is not perfectly uniform. Testable via void ISW measurements.
10. **All Ω values from φ alone** — the formula DM = 1/(φ² + 1 + φ^(−7/2)) uses no fitted parameters. Any future refinement of Planck values either converges toward these predictions or falsifies the formula.

Script Reference

Script	Purpose	Key Result
243BL	Dark energy H(z) test	Best-fit Ωm=0.247, Ωde 9.8% off from $1-1/\varphi^2$
243BL2	Time-frame flip	Ωdm = Ωde/φ^2 (1.3%), Hubble tension = $1/\varphi^5$ (0.7%)
243BL3	Spatial mapping	φ^2 holds at cosmic avg, breaks locally; DM→lensing, DE→redshift
243BL4	Information³ dark sector	DE→DM→baryons = Datum→Signal→Meaning; DM/b = φ^3.5
243BL5	Space/Time ↔ Light/Dark	ARA_space = φ^(−φ); vertical coupler fails as density ratio
243BL6	Coupled grid	Grid is non-separable; 3-component formula avg Δ = 0.43%

243BL7	Origin of 3.5	7/2 from 4-system golden angle; Manhattan distance in coupling space
243BL8	Meta-ARA	Three pairs stack by octave (span ratio 1.97, 1.6% off); φ couplers inside; full formula avg Δ = 0.77%